

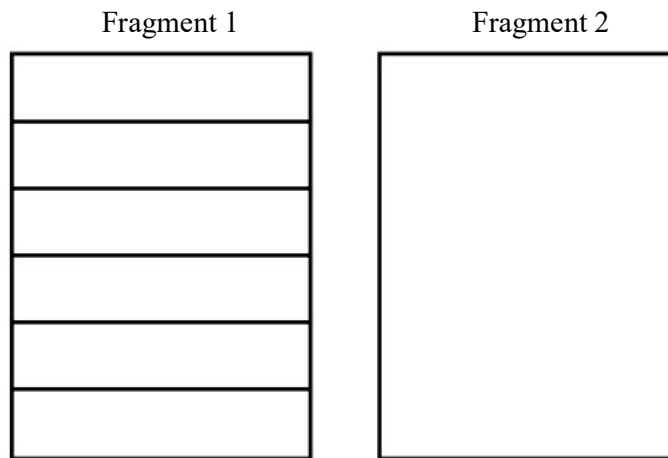
Fragments

A fragment is an Android component that holds part of the behavior and/or UI of an activity. Fragments are not independent entities, and are always embedded in an activity. In many ways, they resemble activities. Think of a fragments as *another form* of activity (“**mini-activities**”). Fragments may be created statically or dynamically.

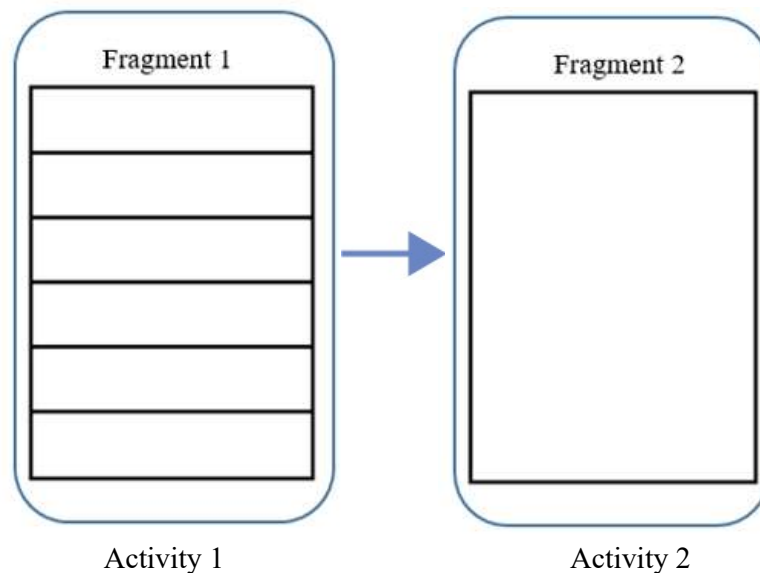
You don’t *have* to use fragments. However, if you do use them and use them well, they can provide:

- *Modularity*: dividing complex activity code across fragments for better organization and maintenance.
- *Reusability*: placing behavior or UI parts into fragments that can be shared across multiple activities.
- *Adaptability*: representing sections of a UI as different fragments and utilizing different layouts depending on screen orientation and size.

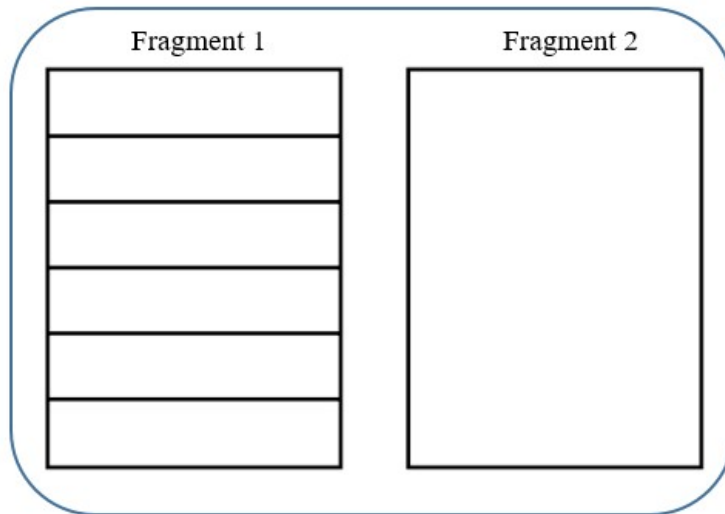
The figure below shows two fragments. Fragment 1 might contain a **List****View** showing a list of movie titles. Fragment 2 might contain some **Text****Views** and **Image****Views** showing some text and images.



Now imagine the application is running on an Android tablet (or on an Android smartphone) in portrait mode. In this case, Fragment 1 might be embedded in one activity, whereas Fragment 2 might be embedded in another activity (see Figure below). When users select an item in the list in Fragment 1, Activity 2 is started.



If the application is now displayed in a tablet in landscape mode, both fragments can be embedded within a single activity, as shown below.



Activity 1

Lab Exercise 1 (*Static Fragments*)



1. Create a new application called *Fragments*
2. Create a new blank fragment called *Fragment1*
3. Replace the text "**hello_blank_fragment**" in *fragment_1.xml* with "**This is fragment #1**"
4. Create a new blank fragment called *Fragment2*
5. Replace the text "**hello_blank_fragment**" in *fragment_2.xml* with "**This is fragment #2**"
6. Modify *activity_main.xml* so that a *LinearLayout* (*vertical*) is used, and delete the **TextView**

7. Drag and drop the **<fragment>** using **Palette** and then **Containers**. Choose **Fragment1**. Insert the attribute *android:layout_weight="1"* inside the fragment tag.
8. Drag and drop the **<fragment>** using **Palette** and then **Containers**. Choose **Fragment2**. Insert the attribute *android:layout_weight="1"* inside the fragment tag.
9. The GUI in *Design* should look similar to the app screen given at the beginning of the exercise.
10. Run the app. How does it work?

Lab Exercise 2 (Dynamic Fragments)

For this exercise we will use the application created in *Lab Exercise 1*. The following changes are necessary.

1. Delete the fragment elements in *activity_main.xml*
2. Delete the statement *setContentView...* from *onCreate()* in *MainActivity*, and then insert the following lines of code.

```
FragmentTransaction ft = getSupportFragmentManager().beginTransaction();
```

```
DisplayMetrics display = this.getResources().getDisplayMetrics();
```

```
int width = display.widthPixels;
```

```
int height = display.heightPixels;
```

```
if (width > height) {
```

```
    Fragment1 f1 = new Fragment1();
```

```
    ft.replace(android.R.id.content,f1);
```

```
} else {
```

```
    Fragment2 f2 = new Fragment2();
```

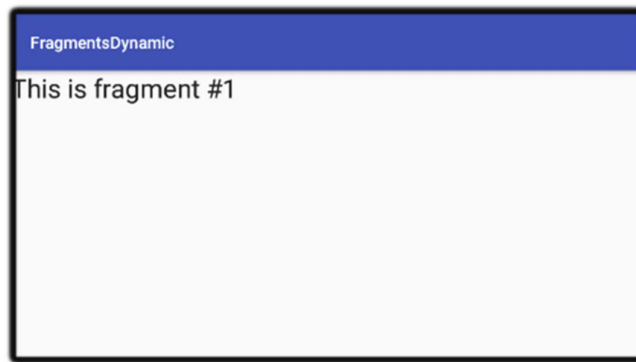
```
    ft.replace(android.R.id.content,f2);
```

```
}
```

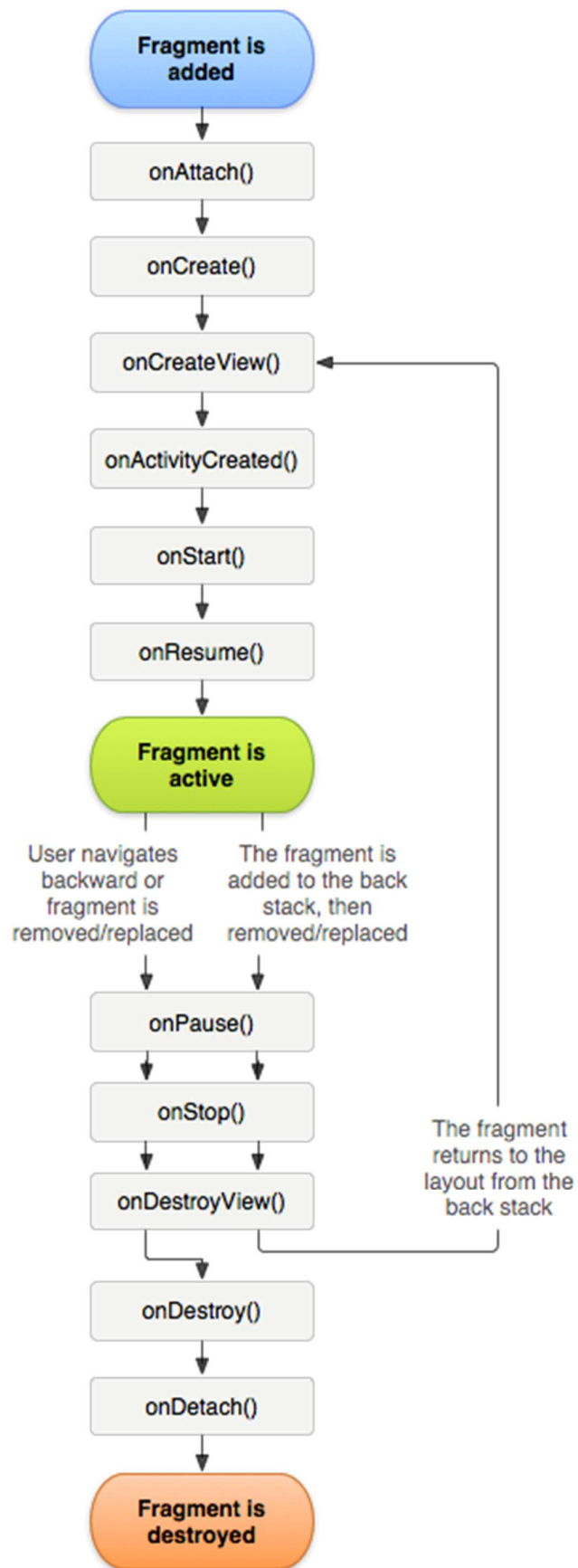
```
ft.addToBackStack(null);
```

```
ft.commit();
```

3. Run the app. How does it work?



Android Fragment Lifecycle



(Source: Android Developer documentation)

The following lifecycle events come into play when you add a fragment:

- **onAttach**: when the fragment attaches to its host activity
- **onCreate**: when a new fragment instance initializes, which always happens after it attaches to the host
- **onCreateView**: when a fragment creates its portion of the view hierarchy, which is added to its activity's view hierarchy
- **onActivityCreated**: when the fragment's activity has finished its own **onCreate** event
- **onStart**: when the fragment is visible; a fragment starts only after its activity starts and often starts immediately after its activity does
- **onResume**: when the fragment is visible and interactable; a fragment resumes only after its activity resumes and often resumes immediately after the activity does
- **onPause**: when the fragment is no longer interactable; this occurs when either the fragment is about to be removed or replaced or when the fragment's activity pauses
- **onStop**: when the fragment is no longer visible; this occurs either after the fragment is about to be removed or replaced or when the fragment's activity stops
- **onDestroyView**: when the view and related resources created in **onCreateView** are removed from the activity's view hierarchy and destroyed
- **onDestroy**: when the fragment does its final clean up
- **onDetach**: when the fragment is detached from its activity

Lab Exercise 3 (Fragment Lifecycle)

1. Create a new application called *FragmentLifeCycle*
2. Delete *setContentView....* from *onCreate()* in *MainActivity*
3. Create a new blank fragment called *Fragment1*
4. Replace the text "**hello_blank_fragment**" in *fragment1.xml* with "**This is fragment #1**"
5. Replace the *Fragment1* class with the code given below.

```
public class Fragment1 extends Fragment {  
  
    @Override  
    public View onCreateView(LayoutInflater inflater, ViewGroup container,  
        Bundle savedInstanceState) {  
  
        Log.d("Fragment 1", "onCreateView");  
  
        return inflater.inflate(R.layout.fragment1, container, false);  
    }  
}
```

```
}
```

```
@Override  
public void onAttach(Context context){  
    super.onAttach(context);  
    Log.d("Fragment 1", "onAttach");  
}
```

```
@Override  
public void onCreate(Bundle savedInstanceState){  
    super.onCreate(savedInstanceState);  
    Log.d("Fragment 1", "onCreate");  
}
```

```
@Override  
public void onActivityCreated(Bundle savedInstanceState){  
    super.onActivityCreated(savedInstanceState);  
    Log.d("Fragment 1", "onActivityCreated");  
}
```

```
@Override  
public void onStart(){  
    super.onStart();  
    Log.d("Fragment 1", "onStart");  
}
```

```
@Override  
public void onResume(){  
    super.onResume();  
    Log.d("Fragment 1", "onResume");  
}
```

```
@Override  
public void onPause(){  
    super.onPause();  
    Log.d("Fragment 1", "onPause");  
}
```

```
@Override  
public void onStop(){  
    super.onStop();  
    Log.d("Fragment 1", "onStop");  
}
```

```
@Override  
public void onDestroyView(){  
    super.onDestroyView();  
    Log.d("Fragment 1", "onDestroyView");  
}
```

```

    @Override
    public void onDestroy(){
        super.onDestroy();
        Log.d("Fragment 1", "onDestroy");
    }

    @Override
    public void onDetach(){
        super.onDetach();
        Log.d("Fragment 1", "onDetach");
    }
}

```

When the app is Run 

```

03-14 22:21:52.602 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onAttach
03-14 22:21:52.602 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onCreate
03-14 22:21:52.605 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onCreateView
03-14 22:21:52.624 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onActivityCreated
03-14 22:21:52.624 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onStart
03-14 22:21:52.626 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onResume

```

Pressing  on emulator

```

03-14 22:22:25.286 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onPause
03-14 22:22:25.355 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onStop

```

Relaunching the app

```

03-14 22:22:50.211 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onStart
03-14 22:22:50.212 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onResume

```

Pressing  on emulator

```

03-14 22:22:55.100 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onPause
03-14 22:22:55.101 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onStop
03-14 22:22:55.101 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onDestroyView
03-14 22:22:55.101 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onDestroy
03-14 22:22:55.101 31290-31290/com.example.arvin.fragmentsd D/Fragment 1: onDetach

```

Interactions between Fragments

Lab Exercise 4 (Communication between Fragments)

You may modify a previous project, or create a new project by following the steps given below.

1. Create a new application called *FragmentsComm*
2. Create a new blank fragment called *Fragment1*
3. Replace the text "**hello_blank_fragment**" in *fragment1.xml* with "**This is fragment #1**"
4. Insert the statement `android:id="@+id/tvFrag1"` in the **TextView** element in *fragment1.xml*

5. Create a new blank fragment called *Fragment2*
6. Insert the following code after the *onCreateView()* method in *Fragment2.java*

```
@Override
public void onStart() {
    super.onStart();
    final Button b = (Button) getActivity().findViewById(R.id.buttonFrag2);

    b.setOnClickListener(new View.OnClickListener() {
        public void onClick(View v) {
            TextView tv = (TextView) getActivity().findViewById(R.id.tvFrag1);
            Toast.makeText(getActivity(), tv.getText(), Toast.LENGTH_SHORT).show();
        }
    });
}
```

7. Modify *fragment_2.xml* so that a *LinearLayout* (*vertical*) is used.
8. Replace the text "**hello_blank_fragment**" in *fragment_2.xml* with "**This is fragment #2**"
9. Use "*wrap_content*" for *android:layout_height* in the **TextView** element in *fragment_2.xml*
10. Create a **Button** below the **TextView** in *fragment_2.xml*. The **Button** element must have the following *xml* code

```
<Button
    android:text="Get text in Fragment #1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/buttonFrag2"
    android:onClick="onClick"/>
```

11. Modify *activity_main.xml* so that a *LinearLayout* (*vertical*) is used, and delete the **TextView**
12. Drag and drop the **<fragment>** using **Palette** and then **Containers**. Choose *Fragment1*. Insert the attribute *android:layout_weight="1"* and replace "*wrap_content*" with "*match_parent*" inside the fragment tag.
13. Drag and drop the **<fragment>** using **Palette** and then **Containers**. Choose *Fragment2*. Insert the attribute *android:layout_weight="1"* and replace "*wrap_content*" with "*match_parent*" inside the fragment tag.
14. Run the app. How does it work?

